

Charles Hung

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EDUCATION

Imperial College London

London, UK

Master of Engineering (MEng), Mechanical Engineering with Nuclear Engineering

Sep 2024 – Present

- Year 1: First Class Honors, Dean's List (2nd); Year 2: First Class Honors, Dean's List (3rd)
- Relevant Coursework: Nuclear Fusion, Thermodynamics, Fluid Mechanics, Materials, Numerical Methods and Analysis, Stress Analysis, Mechanics, Dynamics, Heat Transfer, Mechatronics, Design and Manufacturing

EXPERIENCE

Imperial College AUV Society - Co-Founder and Technical Lead

London, UK

Student-led AUV and Marine Technology Society

May 2026 – Present

- Co-founded and co-chair Imperial's first AUV society, promoting marine technology, R&D, and underwater exploration; leading sponsorship discussions with RS and the college department
- Lead technical direction and mechanical design of the society's first AUV platform.

Zelostech - Chassis Development Intern

Suzhou, China

Chassis Development Team

Mar 2026 – Apr 2026

- Designed and optimized steering-module cable management systems in CATIA for mass production.
- Modeled vehicle dynamics in Simulink, implementing PID controllers and filters to improve system accuracy.
- Ran diagnostics on the steering module via TSMaster and CAN bus to verify hardware-software integration.
- Collaborated with automotive and smart-steering suppliers to support development.

Imperial Racing Green - Mechanical Engineer

London, UK

Imperial Formula Racing (IFR) - Chassis Team 2025

Oct 2025 – Jun 2026

- Designed, manufactured, and tested vehicle chassis components in compliance with FSUK regulations, emphasizing ergonomic and performance-oriented design.
- Optimizing clutch and shifter systems by developing a hand-operated clutch to reduce weight and enhance driver control and comfort.

Imperial College Space Society (ICSS) - Injector Design Engineer

London, UK

Kepler Engine - Rocket Injector Design

Oct 2024 – Jun 2026

- Designed and tested an unlike impinging rocket injector, implemented for a cryogenic, regeneratively-cooled, bipropellant rocket engine, running on isopropyl alcohol (IPA) and liquid oxygen (LOx).
- Utilized CAD and CFD software to analyze injector characteristics and provide feedback for modifications.

Fish Logic Underwater Robotics - Chief Mechanical Engineer

Macau SAR

Siren - Research, Design, Development

June 2022 – June 2024

- Led the mechanical team, aligning functionality, operations, and safety with team design philosophy.
- Developed and integrated innovative ROV tools that addressed mission challenges and optimized tool design, and manufacturing, driving the team to a 2nd-place ranking in the MATE ROV Explorer World Championship.
- Oversaw technical functions, ensuring timely, high-quality deliverables and mentoring younger team members.

PROJECTS

Ion Thruster

Macau SAR

Small-scale Biefeld-Brown Effect Ion Thruster Prototype

2026 – Present

- Constructed a lightweight frame and high-voltage circuit to generate ionic wind.
- Managed all aspects of the hands-on build, optimizing and maximizing thrust, and reiterating the prototypes.

SKILLS AND INTERESTS

Language Skills: Fluent in English and Mandarin.

Tools: Autodesk Inventor, Fusion 360, Onshape, SOLIDWORKS, CATIA, ANSYS FLuent, ANSYS Mechanical, OpenRocket, MATLAB, TSMaster, FDM and SLA 3D printing, Machining, Github, Microsoft Office 365

Technical Skills: Rapid Product Development & Prototyping, Hardware Testing/Data Driven Design, Mechanical Design & Analysis, Hardware Firmware Diagnostics, Computational Modeling, Leadership & Collaboration

Programming Languages: Python, C++, Java, MATLAB